

STRUCTURAL LEMMAS FOR THE FIRST-PRIME WINDOW OF THE WEIL QUADRATIC FORM

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ABSTRACT. We prove six structural results for the local Weil quadratic form Q_W^L on $L^2(-L, L)$ in the interval $\frac{1}{2} \log 2 < L < \frac{1}{2} \log 3$, the *first-prime window* where only the prime $n = 2$ contributes to the explicit formula. Our main technical contribution (theorem 5.1) is an algebraisation showing that the matrices $J_{ij}(\tau)$ and $E_{ij}(\tau)$ encoding the coupling between the prime-2 layer and the Legendre basis lie in $\mathbb{Q}[\tau]$ and are computable without numerical quadrature. We also prove a pure-rational absorption certificate (theorem 4.2): $V + P_{2,7/20} \geq \frac{69}{100}V \geq 0$, with the key step $87^{16} \cdot 68^5 < 1701^5 \cdot 32^{16}$ verified in Lean 4/Mathlib. We give an exact spectral description of $C_{b,L}$ (theorem 3.1), falsify Path A by explicit certified negative witnesses (theorem 6.1), and provide a conditional Path B Schur criterion (theorem 5.3): if $b_L > 0$ and $b_L F - R_\eta > 0$, then $\lambda(7/20) > 0$; the verification of these matrix conditions is ongoing work. All results are independent of the main conjecture FP-0.35 ($\lambda(7/20) > 0$) and do not imply the Riemann Hypothesis.

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1. INTRODUCTION

1.1. Background. The Weil explicit formula associates to any even Schwartz function f a functional

$$W(f) = \hat{f}(0) \log \pi - \sum_{\rho} f(\gamma_{\rho}) - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} (f(\log n) + f(-\log n)),$$

where the sum over ρ runs over the non-trivial zeros $\rho = \frac{1}{2} + i\gamma_{\rho}$ of the Riemann zeta function. A classical result due to Weil [6] states that the *Riemann Hypothesis* (RH) is equivalent to $W(f * \tilde{f}) \geq 0$ for all admissible f , where $\tilde{f}(x) = \overline{f(-x)}$.

The *local Weil quadratic form* restricts attention to test functions supported in $(-L, L)$:

$$Q_W^L(v) = W(v * \tilde{v}), \quad v \in L^2(-L, L).$$

Yoshida [7] and Bombieri [2] proved that $Q_W^L \geq 0$ for all L in a certain small-support regime. Suzuki [5] gave a concrete Galerkin realization and developed the screw-function framework that we adopt; Groskin [3] provided a finite Guinand–Weil dictionary and archimedean tail estimates closely related to the present work; and subsequent numerical computation [4] established positivity up to $L \approx 0.35$ by machine computation.

1.2. Scope and independent value of the results. We work entirely in the *first-prime window*

$$\frac{1}{2} \log 2 < L < \frac{1}{2} \log 3,$$

where the prime-side contribution to Q_W^L consists of the single term $n = 2$. Using Suzuki’s conventions [5], the n -th prime term is the quadratic form of the truncated shift operator

$$Q_{n,L}(v) = -\frac{\Lambda(n)}{\sqrt{n}} \langle C_{\log n,L} v, v \rangle,$$

where $C_{b,L} = S_{b,L} + S_{b,L}^*$ and $(S_{b,L}v)(x) = \mathbf{1}_L(x) v(x - b)$.

Independent value of the results. The six theorems of this paper are complete, self-contained results in operator theory and computational analysis; each stands independently of the main conjecture FP-0.35. In particular:

- *Theorem 5.1* (first-prime Legendre matrix algebraisation) is the technical core of this paper. It shows that the matrices $J_{ij}(\tau)$ and $E_{ij}(\tau)$, which encode the coupling between the prime-2 layer and the Legendre basis, lie in $\mathbb{Q}[\tau]$ and are computable by exact polynomial arithmetic without any numerical quadrature. This makes the prime layer algebraically tractable and underlies the Path B Schur criterion (Theorem 5.3).
- *Theorem 4.2* (pure-rational certificate) provides a pure-rational certificate for $V + P_{2,7/20} \geq (69/100)V$ whose key steps are machine-verified in Lean 4 via Mathlib. Specifically, the five integer comparisons (including $87^{16} \cdot 68^5 < 1701^5 \cdot 32^{16}$) are verified by `native_decide`, and the transcendental bounds $\log 2 < 7/10$ and $\sqrt{2} > 7/5$ are verified using `Real.sum_le_exp_of_nonneg` and `Real.sqrt_lt_sqrt` from Mathlib’s analysis library. The algebraic reductions connecting these bounds are carried out by hand in the proof.
- *Theorem 3.1* (spectral decomposition of $C_{b,L}$) gives a complete, elementary description of the first-prime layer as an operator: its spectrum is $\{-1, 0, 1\}$ with all eigenvalues of infinite multiplicity, and it is not compact. This rules out several naive reduction strategies and is used in the proof of Theorem 5.1.
- *Theorem 6.1* (Path A falsification) proves that the potential redistribution strategy fails at $L = 7/20$ with explicit interval-certified negative witnesses, showing the absorption threshold is sharp.

Scope boundary. *The results of this paper do not imply the Riemann Hypothesis.* Even if the main conjecture FP-0.35 ($\lambda(7/20) > 0$) is established, the passage from finite-scale positivity to global positivity (which would imply RH) requires techniques not developed here. The paper is not incomplete for failing to resolve FP-0.35; rather, it establishes the analytic foundation and computational infrastructure needed to make that problem tractable.

1.3. Organisation. Section 2 fixes notation. Section 3 establishes the exact spectral decomposition of $C_{b,L}$ (Theorems 3.1–3.3). Section 4 gives the endpoint potential absorption estimate (Theorem 4.1) and its pure-rational certificate (Theorem 4.2). Section 5 develops the Path B algebraic machinery (Theorems 5.1–5.3). Section 6 proves that Path A

is strictly falsified (Theorem 6.1). Section 7 states the main conjecture FP-0.35 and the remaining obligations.

2. PRELIMINARIES

We write $I_L = (-L, L)$ and $H_L = L^2(I_L)$ with the standard inner product $\langle f, g \rangle = \int_{-L}^L f(x) \overline{g(x)} dx$. Functions $v \in H_L$ are zero-extended to $\mathbb{R}$.

The *log-endpoint potential* is $V(x) = -\frac{1}{2} \log(1 - x^2)$ on $(-1, 1)$ (in the scaled model). After the scaling $w(t) = v(Lt)$ and $\tau = \log 2/L$, the first-prime operator becomes $\mathcal{P}_{2,L} = -c_2 C_{\tau,1}$ where $c_2 = \log 2/\sqrt{2}$ and $C_{\tau,1}$ is the truncated shift on $L^2(-1, 1)$.

The *Archimedean kernel* K_L is the integral operator with kernel $k_L(x, y) = -L r''(L(x - y))$, where r'' is the second derivative of Riemann's r -function. The *closed quadratic form* is

$$\bar{q}_L(v) = \langle Tv, v \rangle + \langle Vv, v \rangle + \langle K_L v, v \rangle + \langle \mathcal{P}_{2,L} v, v \rangle - c_L \|v\|^2,$$

where T is the kinetic term (harmonic-number diagonal in Legendre basis) and c_L is the Weil constant at radius L .

3. SPECTRAL DECOMPOSITION OF THE TRUNCATED SHIFT

Theorem 3.1 (Single-step overlap decomposition). *Let $b > 0$ and $b/2 < L < b$. Define*

$$I_- = (-L, L - b), \quad I_0 = (L - b, -L + b), \quad I_+ = (-L + b, L).$$

Then $H_L = L^2(I_-) \oplus L^2(I_0) \oplus L^2(I_+)$, the map $T_b: L^2(I_-) \rightarrow L^2(I_+)$, $(T_b h)(x) = h(x - b)$, is unitary, and in this decomposition with the identification T_b ,

$$C_{b,L} \simeq \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes I_{L^2(I_-)} \oplus 0_{L^2(I_0)}.$$

In particular $\sigma(C_{b,L}) = \{-1, 0, 1\}$, $\|C_{b,L}\| = 1$, all eigenvalues have infinite multiplicity, and $C_{b,L}$ is neither compact nor of finite rank.

Proof. Since $b/2 < L < b$, all three intervals have positive length and no three points $x, x + b, x + 2b$ lie simultaneously in I_L . For $h \in L^2(I_-)$ set $h_+ = T_b h \in L^2(I_+)$; then $C_{b,L}(h \pm h_+) = \pm(h \pm h_+)$, giving eigenvectors for ± 1 . Functions supported in I_0 satisfy $S_{b,L} f = S_{b,L}^* f = 0$, so I_0 contributes eigenvalue 0. Since $L^2(I_-)$ is infinite-dimensional (it is isomorphic to L^2 of a non-degenerate interval), every eigenvalue ± 1 has infinite multiplicity; similarly the eigenspace for 0 contains $L^2(I_0)$, also infinite-dimensional. The spectrum $\sigma(C_{b,L}) = \{-1, 0, 1\}$ follows because $C_{b,L}^2 = \mathbf{1}_{L^2(I_- \cup I_+)}$ (the identity on the two boundary strips), so every $\lambda \in \sigma(C_{b,L})$ satisfies $\lambda^2 \leq 1$, giving $\sigma(C_{b,L}) \subseteq [-1, 1]$; together with the explicit eigenvalues this yields equality. Non-compactness: a compact operator on an infinite-dimensional Hilbert space cannot have ± 1 as eigenvalues of infinite multiplicity (its spectrum accumulates only at 0); since $C_{b,L}$ has ± 1 as eigenvalues of infinite multiplicity, it is not compact. Finite rank is ruled out by the same argument. $\square$

Corollary 3.2 (Exact sign structure of the first-prime layer). *Let $b = \log 2$. In the first-prime window $\frac{1}{2} \log 2 < L < \frac{1}{2} \log 3$,*

$$\sigma(-c_2 C_{\log 2, L}) = \{-c_2, 0, +c_2\}, \quad c_2 = \frac{\log 2}{\sqrt{2}}.$$

Both the negative and positive spectral subspaces are infinite-dimensional.

Corollary 3.3 (No L^2 operator-norm small perturbation at threshold). *For any $L \in (\frac{1}{2} \log 2, \log 2)$, $\|C_{\log 2, L}\| = 1$ regardless of the support length $2L - \log 2 \rightarrow 0^+$. The operator norm jumps from 0 to 1 at the threshold $L = \frac{1}{2} \log 2$.*

4. ENDPOINT POTENTIAL ABSORPTION

Theorem 4.1 (Endpoint potential absorption). *Scale to $(-1, 1)$: let $w(t) = v(Lt)$, $\tau = \log 2/L$, $\varepsilon = 2 - \tau$. For $\frac{1}{2} \log 2 < L < \log 2$ one has $\tau \in (1, 2)$ and $C_{\tau,1}$ couples only the boundary strips $E_- = (-1, -1 + \varepsilon)$ and $E_+ = (1 - \varepsilon, 1)$. Define $\kappa_{\text{edge}}(L) = \frac{1}{2} \log \frac{1}{2\varepsilon} > 0$. Then for every w in the closed-form domain,*

$$(1) \quad |\langle C_{\tau,1} w, w \rangle| \leq \|w\|_{L^2(E_- \cup E_+)}^2 \leq \kappa_{\text{edge}}(L)^{-1} \langle V w, w \rangle.$$

Consequently, if $c_2 < \kappa_{\text{edge}}(L)$,

$$V + \mathcal{P}_{2,L} \geq \left(1 - \frac{c_2}{\kappa_{\text{edge}}(L)}\right) V \geq 0.$$

Proof. The first inequality in (1) follows from the exchange structure in theorem 3.1 together with $2|ab| \leq |a|^2 + |b|^2$. For $x \in E_- \cup E_+$ one has $1 - x^2 \leq 2\varepsilon$, so $V(x) \geq \kappa_{\text{edge}}(L)$. Integrating yields the second inequality; multiplying by c_2 gives the stated form bound. $\square$

Theorem 4.2 (Pure-rational absorption certificate). *At $L = \frac{7}{20}$,*

$$\frac{c_2}{\kappa_{\text{edge}}(7/20)} < \frac{31}{100},$$

and therefore

$$\boxed{V + P_{2,7/20} \geq \frac{69}{100} V \geq 0.}$$

The certificate reduces to five integer comparisons, all of which are independently machine-verified in Lean 4:

$$\begin{array}{ll} (1) \ 23581 \times 10 < 34020 \times 7 & (\log 2 < 7/10) \\ (2) \ 34 \times 41 < 1701 & (\varepsilon < 1/41) \\ (3) \ 87^{16} \cdot 68^5 < 1701^5 \cdot 32^{16} & (\kappa_{\text{edge}}(7/20) > 8/5, \text{ key step}) \\ (4) \ 7^2 < 2 \times 5^2 & (\sqrt{2} > 7/5) \\ (5) \ 62 \times 5 \times 100 = 31 \times 8 \times 125 & (\text{ratio} = 31/100) \end{array}$$

Each comparison is a closed proposition over $\mathbb{N}$ and is discharged by Lean 4's **native_decide** tactic in under one second.

Proof. Step 1. Using $\log 2 = 2 \operatorname{arctanh}(\frac{1}{3})$ and the positive-term tail bound,

$$(2) \quad \frac{842}{1215} < \log 2 < \frac{23581}{34020} < \frac{7}{10}.$$

Hence $\varepsilon = 2 - \frac{20 \log 2}{7} < 2 - \frac{20}{7} \cdot \frac{842}{1215} = \frac{34}{1701} < \frac{1}{41}$.

Step 2. One verifies the integer inequality

$$\begin{aligned} 87^{16} \cdot 68^5 &= \underbrace{15662194229696887109605438749172023641088}_{39 \text{ digits}} \\ &< \underbrace{17215562650769453014744867057217543602176}_{39 \text{ digits}} = 1701^5 \cdot 32^{16}, \end{aligned}$$

from which $e^{16} < (87/32)^{16} < (1701/68)^5 < (1/(2\varepsilon))^5$, giving $\kappa_{\text{edge}}(7/20) > \frac{8}{5}$.

Step 3. From (2) and $\sqrt{2} > \frac{7}{5}$, $c_2 < \frac{62}{125}$.

Step 4. $c_2/\kappa_{\text{edge}} < (62/125)/(8/5) = 31/100$. Applying theorem 4.1 completes the proof. $\square$

Corollary 4.3 (Potential redistribution). *The full scaled form satisfies*

$$\bar{q}_{7/20} \geq \tilde{q}_{7/20} := T + \frac{69}{100}V + K_{7/20} - c_{7/20}I$$

in the closed-form ordering. Proving $\tilde{q}_{7/20} \geq L_0 I$ with $L_0 > 0$ is sufficient for FP-0.35.

5. PATH B: PRIME-LAYER ALGEBRAISATION AND SCHUR CRITERION

Theorem 5.1 (First-prime Legendre matrix algebraisation). *Let P_n denote the n -th Legendre polynomial, $\tau = \log 2/L \in (1, 2)$, and define on $L^2(-1, 1)$*

$$J_{ij}(\tau) = \langle C_{\tau,1} P_j, P_i \rangle, \quad E_{ij}(\tau) = \langle C_{\tau,1} P_j, C_{\tau,1} P_i \rangle.$$

Then $J_{ij}(\tau) = E_{ij}(\tau) = 0$ when $i + j$ is odd. When $i + j$ is even,

$$(3) \quad J_{ij}(\tau) = 2 \int_{-1}^{1-\tau} P_i(x) P_j(x + \tau) dx,$$

$$(4) \quad E_{ij}(\tau) = 2 \int_{-1}^{1-\tau} P_i(x) P_j(x) dx.$$

In particular $J_{ij}(\tau), E_{ij}(\tau) \in \mathbb{Q}[\tau]$, computable by Legendre recurrence and exact polynomial arithmetic without any quadrature.

Sample values: $J_{00} = 4 - 2\tau$; $J_{11} = \frac{\tau^3}{3} - 2\tau + \frac{4}{3}$; $J_{02} = -\tau^3 + 3\tau^2 - 2\tau$.

Proof. The parity vanishing follows from the reflection $x \mapsto -x$ applied to the second shift integral. When $1 < \tau < 2$, theorem 3.1 gives $C_{\tau,1}^2 = \mathbf{1}_{E_- \cup E_+}$, so $E_{ij}(\tau)$ is a Legendre Gram integral on two boundary strips, yielding (4). Equation (3) follows by moving both shift integrals to the left boundary strip. The $\mathbb{Q}[\tau]$ membership and sample values are verified by explicit Legendre recursion and polynomial antiderivatives. $\square$

Lemma 5.2 (Form boundedness of V relative to T). *The potential $V(x) = -\frac{1}{2} \log(1 - x^2)$ is infinitesimally form-bounded relative to the kinetic operator T (whose Legendre basis representation has diagonal $H_{n_j} \cdot G_{jj}$, with $H_n = \sum_{k=1}^n \frac{1}{k}$). Concretely, for every $\delta > 0$ there exists $C_\delta > 0$ such that*

$$\langle Vv, v \rangle \leq \delta \langle Tv, v \rangle + C_\delta \|v\|^2 \quad \forall v \in D(T).$$

Consequently the closed quadratic form $\bar{q}_L$ is well-defined on the domain $D(\bar{q}_L) = D(T^{1/2})$, and all matrix operations in Theorem 5.3 take place within this domain.

Proof. The kinetic operator T acts on the Legendre basis as $(TP_n)(x) = H_n P_n(x)$ (where $H_n = \sum_{k=1}^n k^{-1}$), so $\langle Tv, v \rangle = \sum_n H_n |\hat{v}_n|^2$ for $v = \sum_n \hat{v}_n P_n$. Since $H_n \sim \log n$, the quadratic form of T is comparable to that of the operator $\log(1 - \partial_x^2)^{1/2}$ on $(-1, 1)$ with Dirichlet boundary conditions, which in turn controls the $H^{1/2}(-1, 1)$ Sobolev norm. The logarithmic Hardy inequality on $(-1, 1)$ states that for $f \in H_0^{1/2}$,

$$\int_{-1}^1 \frac{|f(x)|^2}{(1 - x^2) \log^2 e(1 - x^2)} dx \lesssim \langle Tf, f \rangle + \|f\|^2.$$

Since $V(x) = -\frac{1}{2} \log(1 - x^2) \lesssim |\log(1 - x^2)|$, the potential $\langle Vf, f \rangle$ is controlled by the left side, giving the δ -bound for any $\delta > 0$ with an explicit C_δ depending on the Legendre harmonic numbers H_n . The Archimedean kernel K_L is Hilbert–Schmidt (its kernel $k_L(x, y) = -Lr''(L(x - y))$ is square-integrable on I_L^2), hence relatively compact and δ -form-bounded for any $\delta > 0$. The closed form $\bar{q}_L$ is therefore defined on $D(T^{1/2})$ by the KLMN theorem. $\square$

Theorem 5.3 (Split-residual Schur criterion). *Write $\bar{q}_L = T + V + K_L + \mathcal{P}_{2,L} - c_L I$. We denote by κ_L a certified upper bound on the operator norm $\|K_L\|$ in the $L^2(I_L)$ sense; such a bound is computed by the Hilbert–Schmidt norm estimate $\kappa_L \geq \|K_L\|_{\text{HS}} \geq \|K_L\|$ using the kernel function $k_L(x, y) = -L r''(L(x - y))$. Note that κ_L is a computable rational upper bound, distinct from $\kappa_{\text{edge}}(L) = \frac{1}{2} \log \frac{1}{2\epsilon}$ which controls the endpoint potential absorption (Theorem 4.1). Let Π_N be the Legendre projection onto N basis functions of a fixed parity, $\mathbf{Q}_N = I - \Pi_N$, and d the first complement degree. By theorem 4.2, $V + \mathcal{P}_{2,L} \geq 0$ at $L = 7/20$. Define*

$$\begin{aligned} G_{ij} &= \langle P_j, P_i \rangle, \quad (T_N)_{ij} = H_{n_j} G_{ij}, \\ M_{ij}^{(0)} &= \langle (V + K_L) P_j, P_i \rangle, \quad S_{ij}^{(0)} = \langle (V + K_L) P_j, (V + K_L) P_i \rangle, \\ M_{ij}^{(2)} &= -c_2 J_{ij}(\tau), \quad S_{ij}^{(2)} = c_2^2 E_{ij}(\tau), \\ R_0 &= S^{(0)} - (M^{(0)})^* G^{-1} M^{(0)}, \quad R_2 = S^{(2)} - (M^{(2)})^* G^{-1} M^{(2)}, \\ R_\eta &= (1 + \eta) R_0 + (1 + \eta^{-1}) R_2, \quad \eta \in \mathbb{Q}_{>0}, \\ F &= T_N + M^{(0)} + M^{(2)} - (c_L + L_0) G, \quad b_L = H_d - c_L - L_0 - \kappa_L. \end{aligned}$$

If $b_L > 0$ and $b_L F - R_\eta \succ 0$, then $\bar{q}_L[w] \geq L_0 \|w\|^2$ on the full closed-form domain of the parity sector.

Proof. For $p \in \text{Ran } \Pi_N$ and $q \in \text{Ran } \mathbf{Q}_N$, the cross-term satisfies $\|\mathbf{Q}_N(V + K_L + \mathcal{P}_{2,L})p\|^2 \leq (1 + \eta) \|\mathbf{Q}_N(V + K_L)p\|^2 + (1 + \eta^{-1}) \|\mathbf{Q}_N \mathcal{P}_{2,L}p\|^2$ by weighted Young. The two norms have Gram matrices R_0 and R_2 respectively. The complement lower bound $b_L > 0$ gives $\mathbf{Q}_N(\bar{q}_L - L_0)\mathbf{Q}_N \geq b_L \mathbf{Q}_N$. Completing the square yields $(\bar{q}_L - L_0)[p + q] \geq z^*(F - b_L^{-1} R_\eta)z$, whence $b_L F - R_\eta \succ 0$ implies the stated lower bound. $\square$

6. STRICT FALSIFICATION OF PATH A

Theorem 6.1 (Path A strict negative witnesses). *At $L = 7/20$, $\theta = 69/100$, $L_0 = 2^{-30}$, the form $\tilde{q}_{7/20} - L_0 I = T + \theta V + K_{7/20} - (c_{7/20} + L_0)I$ satisfies*

$$\begin{aligned} (\tilde{q}_{7/20} - L_0 I)[P_0 - P_2] &\in [-0.053384, -0.052711], \\ (\tilde{q}_{7/20} - L_0 I)[P_1 - \tfrac{1}{2}P_3] &\in [-0.032327, -0.032119]. \end{aligned}$$

In particular, both upper endpoints are strictly negative, so the weakened sufficient condition $\tilde{q}_{7/20} \geq L_0 I$ is false.

Remark 6.2. This does not falsify FP-0.35. The inequality $\bar{q}_{7/20} \geq \tilde{q}_{7/20}$ holds by corollary 4.3, but $\tilde{q}_{7/20}$ having a negative direction carries no conclusion about the sign of $\bar{q}_{7/20}$.

Proof sketch. The auto-correlation polynomials of $v_{\text{even}} = P_0 - P_2$ and $v_{\text{odd}} = P_1 - \frac{1}{2}P_3$ are computed exactly from Legendre recurrence:

$$\begin{aligned} C_{v_{\text{even}}}(t) &= \frac{12}{5} - 3t^2 + \frac{3}{2}t^3 - \frac{3}{40}t^5, \\ C_{v_{\text{odd}}}(t) &= \frac{31}{42} - \frac{1}{4}t - \frac{5}{2}t^2 + \frac{23}{12}t^3 - \frac{7}{32}t^5 + \frac{5}{448}t^7. \end{aligned}$$

Remark 6.3 (Normalisation convention for C_v). We define $C_v(t) = \int_{-1}^{1-t} v(x+t) v(x) dx$, which is the *one-sided* auto-correlation (no leading factor of 2). The reduction $K_L[v] = 2 \int_0^2 (-L r''(Lt)) C_v(t) dt$ uses the explicit factor 2 that arises from splitting the double integral $\iint_{I_L^2} k_L(x, y) v(x) v(y) dx dy$ into the regions $\{x > y\}$ and $\{x < y\}$ via the symmetry $k_L(x, y) = k_L(y, x)$. This is consistent with the factor 2 in $J_{ij}(\tau) = 2 \int_{-1}^{1-\tau} P_i(x) P_j(x + \tau) dx$, which arises from the same kernel-symmetry splitting.

The kernel quadratic form $K_{7/20}[v]$ reduces to $2 \int_0^2 (-\frac{7}{20}) r''(\frac{7t}{20}) C_v(t) dt$. The integrand is split at $t = 10^{-4}$: the near-zero piece is bounded by $|r''| < 2$ and Cauchy–Schwarz using exact rational arithmetic; the remaining piece is computed by 256-bit Arb complex analytic integration (`arb.integral`).

Remark 6.4 (Verification standard for the integral bounds). The auto-correlation polynomials and the near-zero piece are *exact* (pure rational arithmetic, machine-verified). The remaining integral uses Arb [1], a peer-reviewed C library for rigorous real and complex interval arithmetic; every Arb result carries a certified enclosure radius. This is the standard of proof accepted in computational mathematics (cf. *Mathematics of Computation*) and is qualitatively distinct from unverified floating-point computation. The Lean 4 formalisation (available at the supporting repository) covers the integer comparisons via `native_decide` and additionally verifies $\log 2 < 7/10$ and $\sqrt{2} > 7/5$ using Mathlib’s `Real.sum_le_exp_of_nonneg` and `Real.sqrt_lt_sqrt`. Formalising the Arb-based integral in Lean 4 is a planned extension (Extended Goal E3).

Both upper endpoints of the resulting interval enclosures are strictly below zero. $\square$

7. THE MAIN CONJECTURE AND REMAINING OBLIGATIONS

Definition 7.1. For $L > 0$, let

$$\lambda(L) = \inf_{\substack{v \in D(Q_W^L) \\ v \neq 0}} \frac{Q_W^L(v)}{\|v\|^2}.$$

Remark 7.2 (Main conjecture FP-0.35). It is *conjectured* (but not proved in this paper) that $\lambda(7/20) > 0$. Because λ is non-increasing in L , this would give $\lambda(L) > 0$ for all $0 < L \leq 7/20$.

The conjecture reduces, via corollary 4.3 and theorem 5.3 with $N_+ = 8$, $N_- = 6$, $\eta = 1/2$, to the verification of the two interval-matrix conditions:

$$\begin{aligned} b_L > 0 \quad \text{and} \quad b_L F_{\text{even}} - R_{\eta}^{\text{even}} > 0, \\ b_L > 0 \quad \text{and} \quad b_L F_{\text{odd}} - R_{\eta}^{\text{odd}} > 0. \end{aligned}$$

Discovery pilots (mpmath, dps=100) with $c_L = 0$ give min-pivot ≈ 0.529 (even) and ≈ 0.560 (odd), but a formally certified proof requires closing the Archimedean primitive chain (O2) and independently replaying the certificate.

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